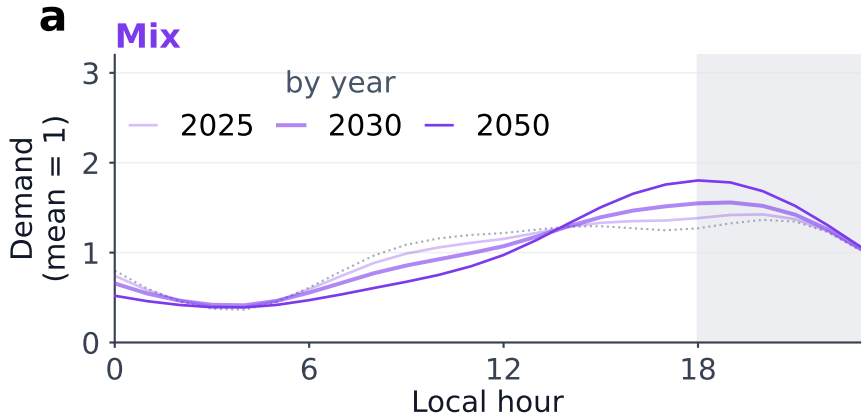
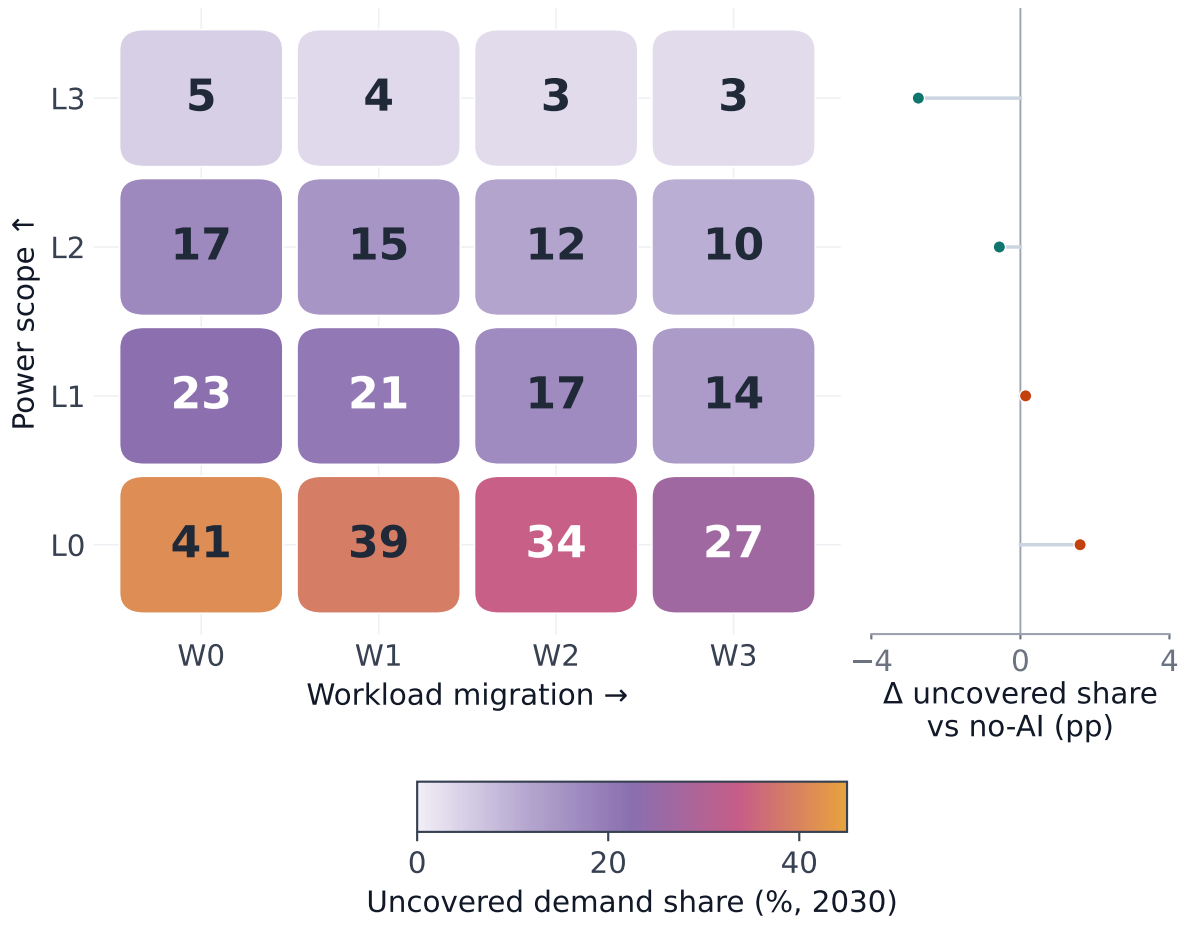
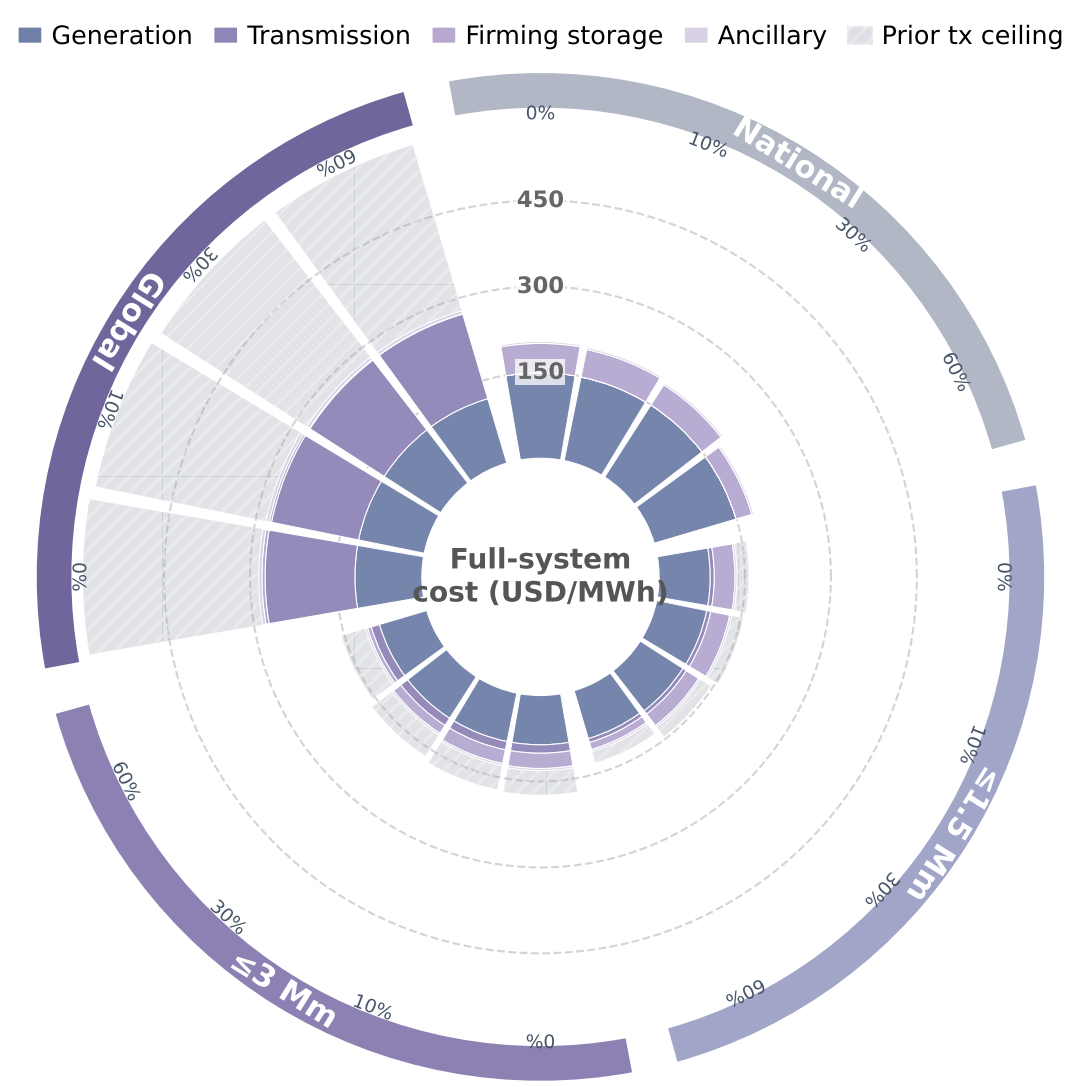
a

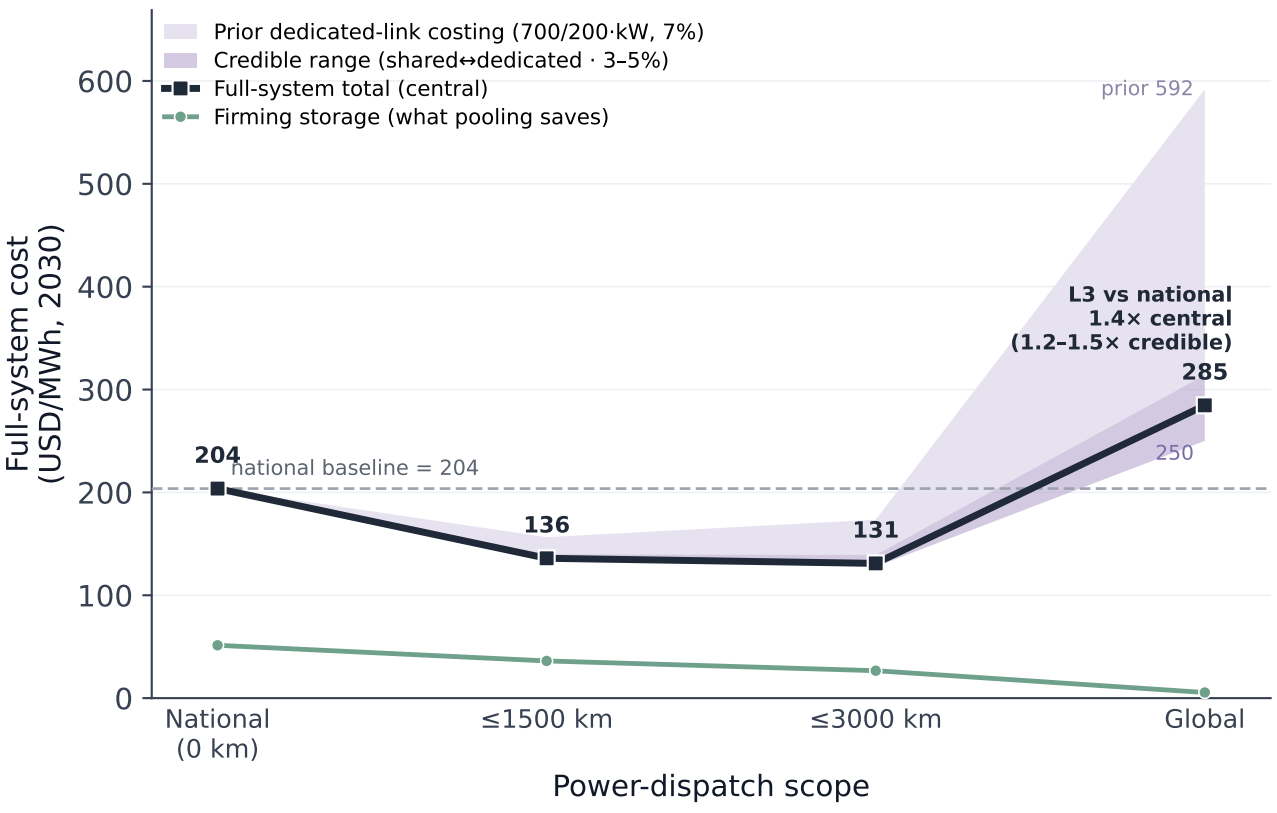

Mix
by year
2025
2030
2050
Demand (mean = 1)
Local hour


L3
L2
L1
L0
Power scope ↑
5
4
3
3
17
15
12
10
23
21
17
14
41
39
34
27
W0
W1
W2
W3
Workload migration →
−4
0
4
Δ uncovered share vs no-AI (pp)
0
20
40
Uncovered demand share (%, 2030)

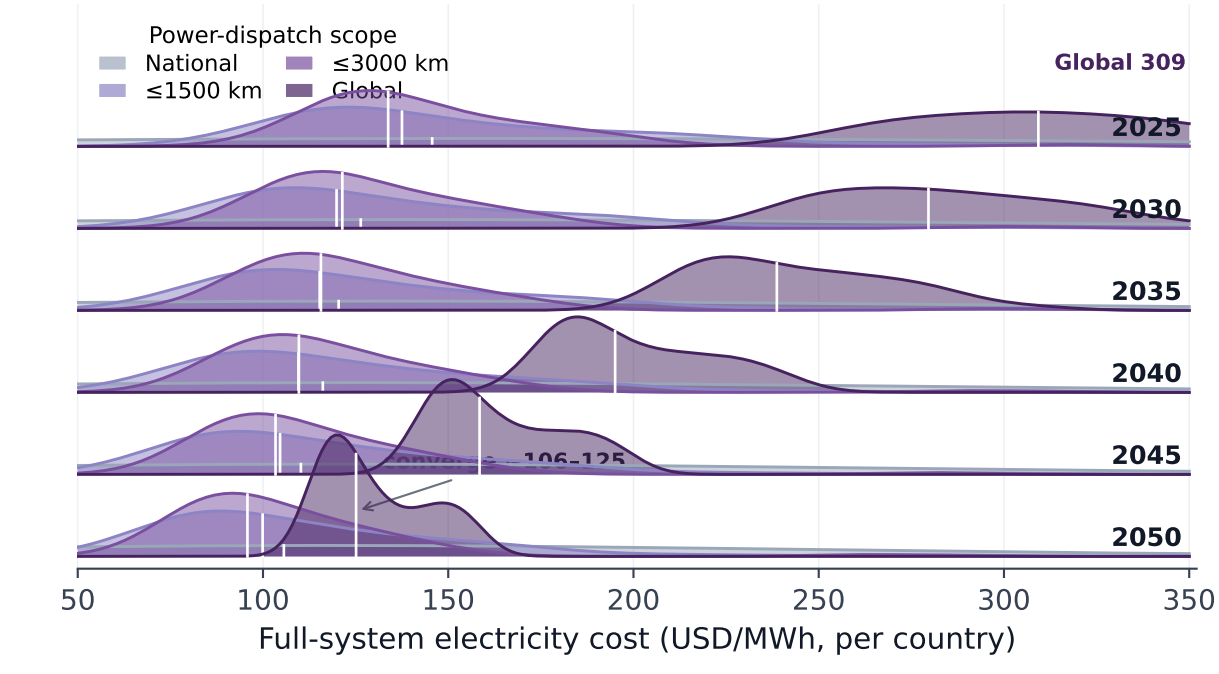
b


Generation
Transmission
Firming storage
Ancillary
Prior tx ceiling
National
Global
≤1,5 Mm
≤3 Mm
0%
10%
30%
60%
450
300
150
Full-system cost (USD/MWh)

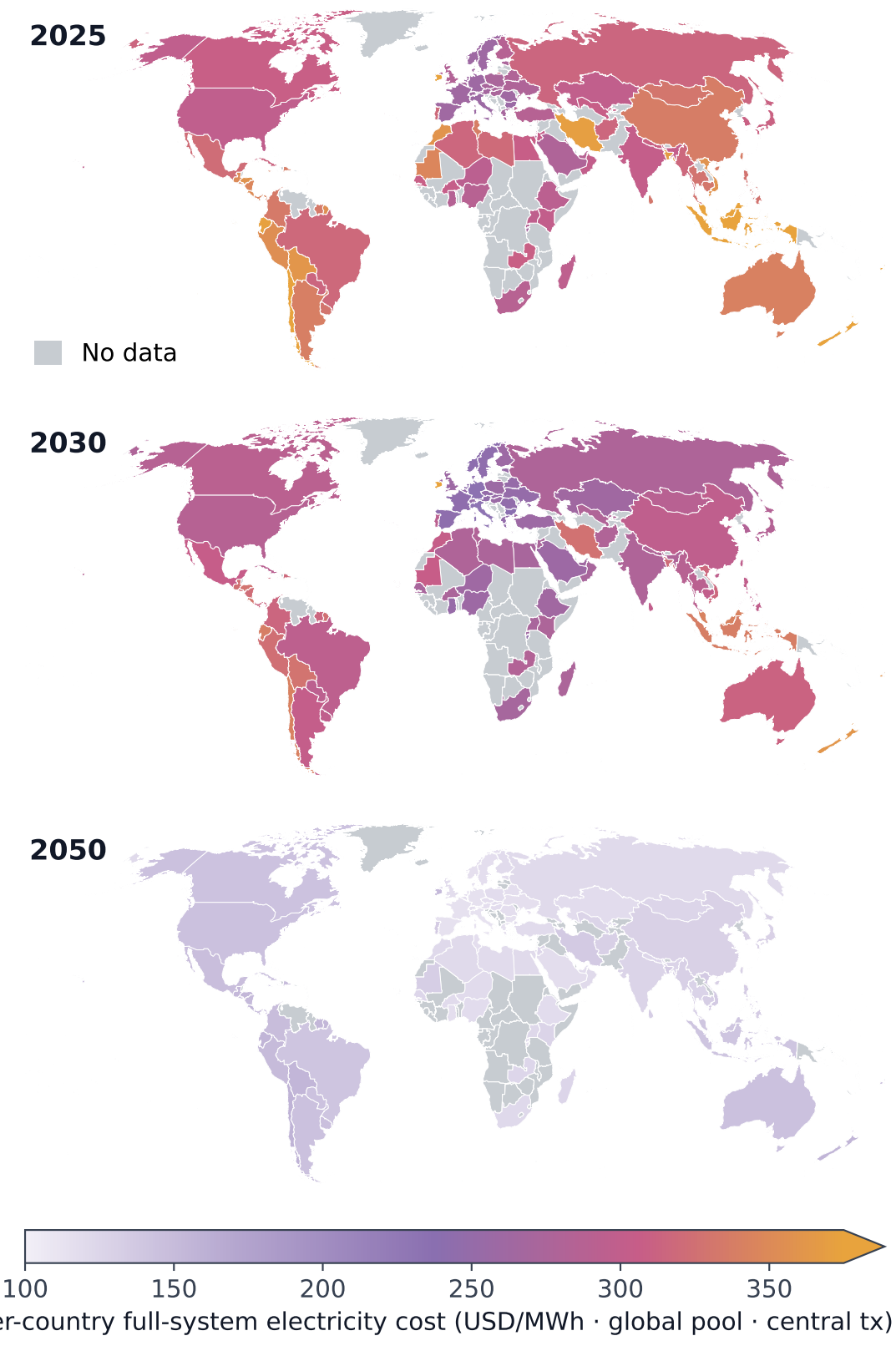
c


Prior dedicated-link costing (700/200·kW, 7%)
Credible range (shared↔dedicated · 3–5%)
Full-system total (central)
Firming storage (what pooling saves)
Full-system cost (USD/MWh, 2030)
prior 592
L3 vs national
1.4× central
(1.2–1.5× credible)
285
250
204
national baseline = 204
136
131
National (0 km)
≤1500 km
≤3000 km
Global
Power-dispatch scope

d


Power-dispatch scope
National
≤1500 km
≤3000 km
Global
Global 309
2025
2030
2035
2040
2045
2050
50
100
150
200
250
300
350
Full-system electricity cost (USD/MWh, per country)

e


2025
No data
2030
2050
100
150
200
250
300
350
Per-country full-system electricity cost (USD/MWh · global pool · central tx)